MECHATRONICS AND MEMS

Subject Code : 23A03702d

UNIT II : Sensors

Syllabus: Static and dynamic characteristics of sensors, Displacement, Position and Proximity sensors, Force and torque sensors, Pressure sensors, Flow sensors, Temperature sensors, Acceleration sensors, Level sensors, Light sensors, Smart material sensors, Micro and Nano sensors, Selection criteria for sensors.

I. Introduction to Sensors

- A **sensor** is a device that detects a physical quantity (measurand) such as displacement, force, temperature, flow, and gives an output signal, usually electrical, proportional to it.
- A **transducer** converts one form of energy to another. Sensor = sensing element; transducer = sensing element + signal conversion. (In practice both words are used for each other.)
- **Classification**
  - **Active sensors** (self generating): give voltage or current without external power. Eg: thermocouple, piezoelectric, photovoltaic.
  - **Passive sensors:** need external excitation; they change R, L or C. Eg: strain gauge, RTD, LVDT, thermistor.
  - **Analog sensors** give continuous output (LVDT, potentiometer); **digital sensors** give pulses or codes (encoder).
  - By measurand: displacement, force, pressure, flow, temperature, level,

light, acceleration etc.

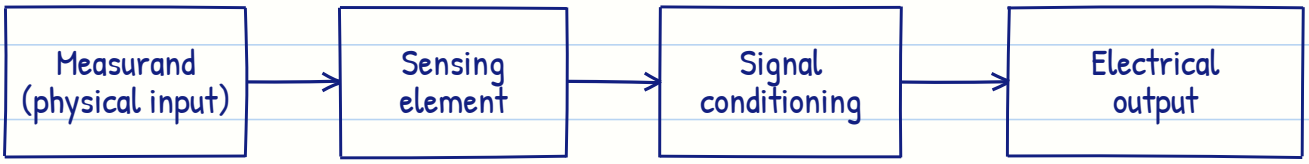


Fig: Sensor - basic block diagram

2. Static Characteristics of Sensors

Static characteristics describe the output when the input is steady or changes very slowly.

- **Range:** minimum to maximum value of input that can be measured. Eg: 0 to 100 deg C.
- **Span:** difference of maximum and minimum of the range. Span = Xmax - Xmin.
- **Accuracy:** closeness of measured value to the true value. Expressed as % of full scale (FS) or % of reading.
- **Precision:** closeness of repeated readings with each other (it is not the same as accuracy).
- **Sensitivity:** change in output per unit change in input = dy/dx. Eg: 10 mV per deg C.
- **Resolution:** the smallest change in input which the sensor can detect.
- **Threshold:** smallest input needed to get any output from zero.
- **Linearity:** how closely the output-input curve is a straight line. Non-linearity = max deviation from best straight line / full scale

output x 100.

- **Hysteresis:** difference in output for the same input when approached from increasing and decreasing directions.
- **Repeatability:** same output for same input under same conditions, in same direction.
- **Reproducibility:** same result when measured by different persons / places / time.
- **Drift:** slow change of output with time at constant input (due to ageing, temperature).
- **Zero error / Offset:** non-zero output at zero input.
- **Dead band (dead zone):** range of input where there is no output change.
- **Stability:** ability to keep its performance over a long time.
- **Error** = Measured value - True value.

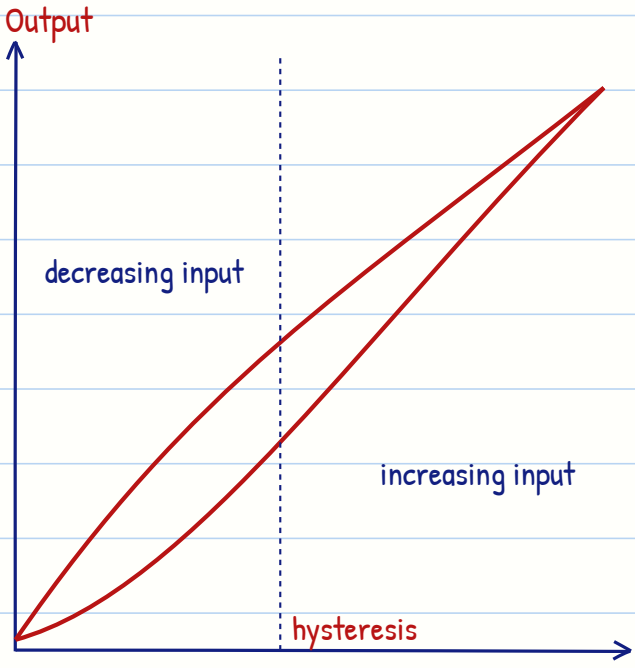


Fig: Hysteresis - different paths for increasing and decreasing input

Accuracy vs Precision (target idea)

- Accurate and precise: all shots at the bullseye. Precise but not accurate: shots grouped but away from centre. Accurate but not precise: shots scattered around the centre.

Calibration

- Comparing the sensor output with a standard of higher accuracy and correcting it. Done periodically to remove zero error and drift.

3. Dynamic Characteristics of Sensors

Dynamic characteristics describe how the sensor responds when the input **changes with time** .

- **Speed of response:** how quickly the output follows the input change.
- **Response time:** time taken to reach a given percentage (say 95% or 98%) of the final value after a step input.
- **Rise time:** time to rise from 10% to 90% of final value.
- **Settling time:** time after which output stays within +/- 2% (or 5%) of final value.
- **Time constant (tau):** time to reach **63.2%** of the final value for a first order system.
- **Lag:** delay of output behind input. **Dynamic error:** difference between true and indicated value when input varies.
- **Fidelity:** ability to reproduce the input variations without distortion.
- **Bandwidth:** range of frequencies over which the sensor responds properly (up to -3 dB).
- **Overshoot, damping ratio and natural frequency** in a second order

sensor (accelerometer, pressure).

Order of the system

- **Zero order:** output follows input instantly. y = K x (potentiometer).
- **First order:** one energy storage, tau dy/dt + y = K x. Step response: y = K x (1 - e^(-t/tau)). Eg: thermocouple, RTD in a thermowell.
- **Second order:** two energy storages, oscillatory. Eg: accelerometer, spring mass damper type sensors, diaphragm pressure sensor.

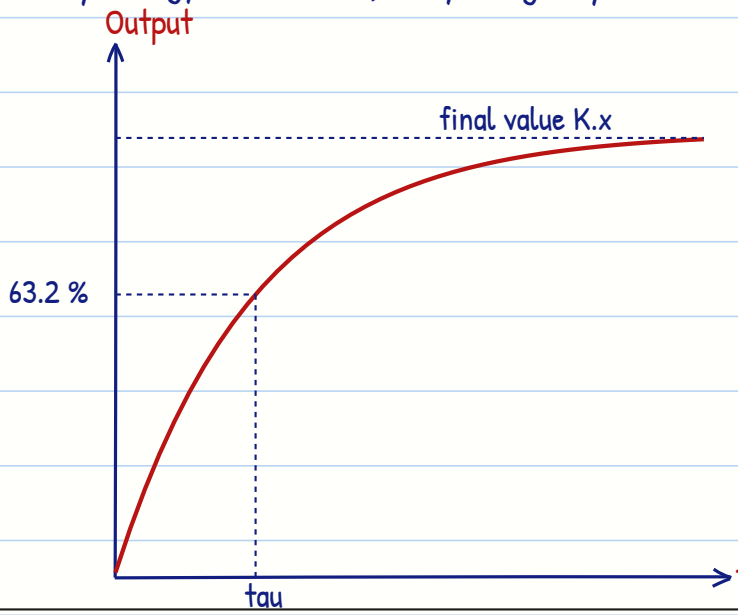


Fig: First order step response - y = K x (1 - e^(-t/tau))

4. Displacement, Position and Proximity Sensors

- **Displacement** = movement of a body from one position to another (linear or angular). **Position** = location of a body w.r.t. a reference. **Proximity** = detecting the presence of an object nearby, without contact.

4.1 Potentiometer (resistive)

- A sliding contact (wiper) moves over a resistive track. Output voltage Vo = Vs x (x / L). Used for linear and rotary displacement.

- Simple and cheap. But friction, wear, noise, limited life and resolution.

4.2 Strain gauge (for small displacement)

- Resistance changes with strain: **Gauge factor GF = (dR/R) / strain** . Used with Wheatstone bridge.

4.3 Capacitive sensor

- Capacitance C = e A / d. Displacement changes the gap d, area A or dielectric. Non contact, high resolution, used for micro displacement and thickness.

4.4 LVDT (Linear Variable Differential Transformer)

- A **primary coil** at the centre and **two secondary coils** wound on either side, connected in series opposition; a movable **ferromagnetic core** inside.
- AC excitation (1 to 10 kHz) in primary. Core at centre: the two secondary voltages are equal and opposite, so Vo = 0 (**null position**).
- Core moves right: V1 > V2, so Vo is proportional to the displacement; **phase** tells the direction (in phase / 180 deg out of phase).
- **Advantages:** frictionless, infinite resolution, high accuracy, robust, range from +/-0.1 mm to +/-250 mm. **Disadvantages:** needs AC supply, demodulator, and is sensitive to stray fields.

Vo = V1 - V2

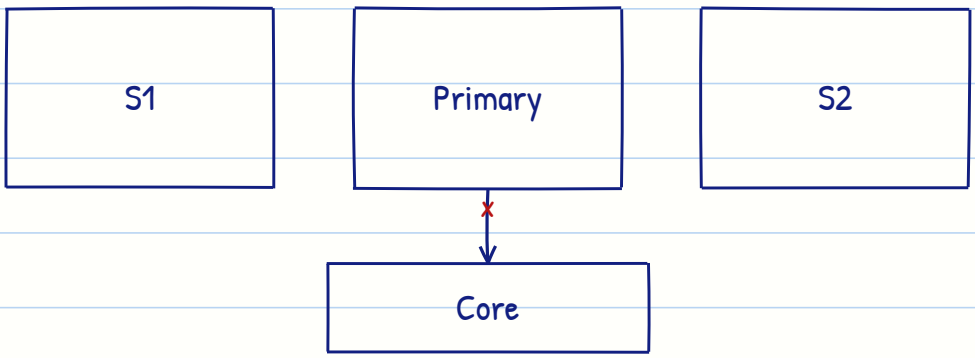


Fig: LVDT - primary, two secondaries and movable core

4.5 Encoders (digital position sensors)

- **Incremental encoder:** a disc with slots, LED and photo detector produce pulses. Counting pulses gives position; two channels A and B (90 deg shifted) give direction; index pulse Z gives reference. Loses position if power fails.
- **Absolute encoder:** disc with coded tracks (binary or **Gray code**) gives a unique code for each position. Position is known directly and is not lost at power off.
- Resolution = 360 / (pulses per revolution). Uses: CNC axes, robot joints, motor speed.

4.6 Resolver and synchro

- Rotating transformers giving sin and cos of shaft angle. Rugged, good for harsh places.

4.7 Proximity sensors

- **Inductive:** detects only **metal** objects; a coil makes an oscillating field,

eddy currents in the metal reduce the oscillation. Range 1 to 30 mm.

- **Capacitive:** detects metals and non metals (plastic, liquid, wood, powder) by change in capacitance.
- **Photoelectric (optical):** LED/laser and photodiode - through beam, retro reflective, diffuse types.
- **Ultrasonic:** transmits sound pulse, measures echo time. Distance = (v x t) / 2.
- **Hall effect:** detects the magnetic field of a magnet. Also **reed switch** (magnetic contact) and limit switches (mechanical).

5. Force and Torque Sensors

5.1 Force sensors

- **Strain gauge load cell:** a metal element (beam, ring, column) deforms under force; gauges bonded on it change resistance. The Wheatstone bridge gives output in mV proportional to the force.
- **Piezoelectric force sensor:** quartz crystal gives charge on loading; good for dynamic force (cutting force) but not static.
- **Capacitive and LVDT type** load cells. **Force sensing resistor (FSR):** resistance falls with force; cheap, used in touch pads and robot grippers.

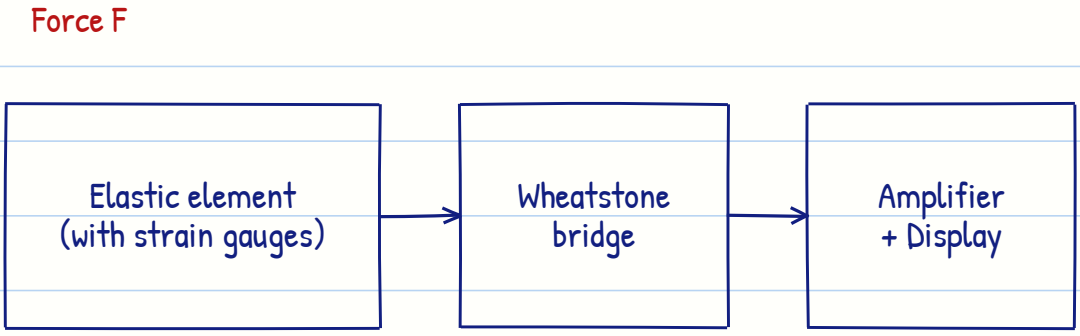


Fig: Strain gauge load cell - block diagram

5.2 Torque sensors

- **Strain gauge torque sensor:** gauges bonded at 45 deg on a shaft measure shear strain due to torsion. Signal taken by slip rings or wireless telemetry.
- **Reaction torque sensor** (static) and **magnetoelastic / optical** non-contact torque sensors.
- Torque T = G J (theta) / L, or from strain: T = (shear strain x G x J) / r.
- Uses: motor testing, robot joints, tightening tools, torque control.

6. Pressure Sensors

- Pressure = Force / Area. Types: absolute, gauge and differential pressure.
- **Elastic types:** diaphragm, **Bourdon tube** , bellows, capsule - the elastic element deflects with pressure. The deflection is converted to an electrical signal by strain gauge, LVDT, capacitive or potentiometer.
- **Strain gauge (piezoresistive) pressure sensor:** gauges on a silicon

diaphragm; widely used, MEMS based.

- **Capacitive pressure sensor:** diaphragm is one plate of a capacitor; pressure changes the gap.
- **Piezoelectric:** quartz - for dynamic pressure, such as in engine cylinders.
- Range: from vacuum to 1000 bar. Used in hydraulic/pneumatic systems, automobile (tyre, MAP), medical (blood pressure).

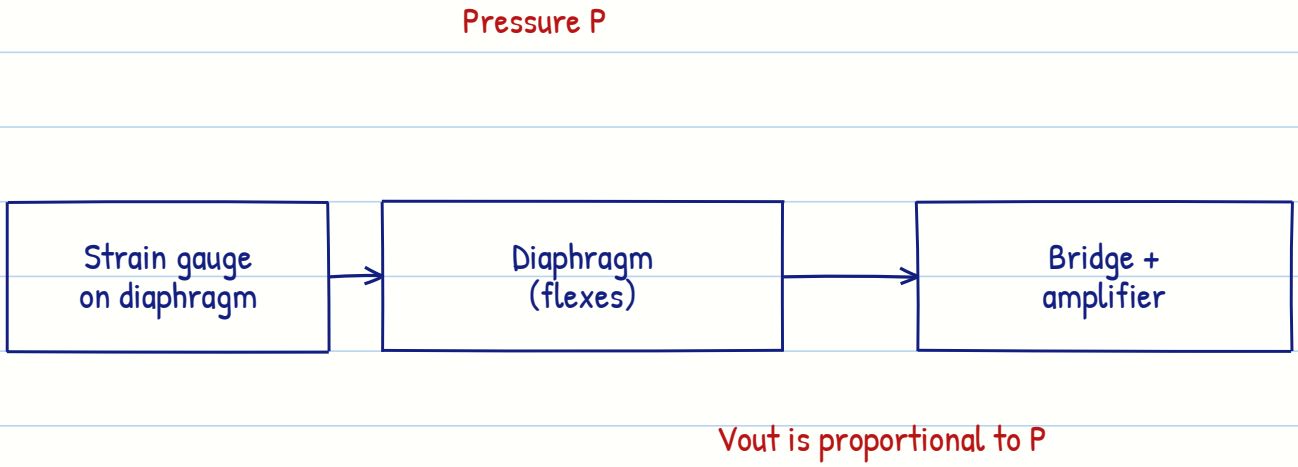


Fig: Diaphragm type pressure sensor

7. Flow Sensors

- Flow rate Q = A x v (m3/s). Used in process plants, fuel systems, water supply and hydraulics.
- **Differential pressure types:** an obstruction like **orifice plate, venturi or flow nozzle** creates a pressure drop. Q is proportional to the square root of the pressure difference (dP). Orifice is cheap, venturi has lower loss.
- **Turbine flow meter:** a rotor spins with speed proportional to flow; a

magnetic pickup gives pulses.

- **Electromagnetic flow meter:** Faraday's law, e = B l v; only for conductive liquids; no obstruction.
- **Ultrasonic flow meter:** transit time or Doppler principle; clamp-on, non invasive.
- **Rotameter** (variable area), **vortex shedding meter** , **coriolis mass flow meter** and **hot wire (thermal) anemometer** for gases.

Type	Principle	Remark
Orifice / Venturi	Pressure drop, Q ~ sqrt(dP)	Cheap, causes pressure loss
Turbine	Rotor speed ~ flow	Good accuracy, clean fluids
Electromagnetic	Faraday's law	Only conducting liquids
Ultrasonic	Transit time / Doppler	Non-invasive, costly

8. Temperature Sensors

- **Thermocouple (Seebeck effect):** two dissimilar metals joined at a hot junction give an emf proportional to the temperature difference with the cold junction. Types: K (Chromel-Alumel, up to 1200 deg C), J (Iron-Constantan), T (Copper-Constantan), R/S (Pt-Rh, up to 1600 deg C). Self generating, wide range, rugged, fast; low voltage so needs cold junction compensation.
- **RTD (Resistance Temperature Detector):** metal wire (platinum **Pt100** , 100 ohm at 0 deg C) whose resistance rises with temperature: R = R0 (1 + alpha T). High accuracy, stable, linear; costly and slower. Range -200 to 650 deg C.

- **Thermistor:** semiconductor (metal oxide); **NTC** type resistance falls strongly with temperature (non linear), **PTC** rises. Very sensitive, cheap, small; limited range (-50 to 150 deg C).
- **IC sensors** (LM35: 10 mV per deg C; AD590) - linear, easy to use. **Bimetallic strip** - mechanical, used in thermostats. **Pyrometer / infrared** - non contact, for very hot or moving objects.

Sensor	Range (deg C)	Output	Features
Thermocouple	-200 to 1700	mV (emf)	Rugged, fast, cheap
RTD (Pt100)	-200 to 650	Resistance change	Accurate, stable
Thermistor	-50 to 150	Resistance change	Very sensitive, non linear
IC (LM35)	-55 to 150	10 mV per deg C	Linear, easy

9. Acceleration Sensors

- Acceleration is measured using a **seismic mass (proof mass)** m on a spring k with damper c inside a case. When the case accelerates, the mass lags and the relative displacement is proportional to acceleration: x = (m a) / k.
- The displacement is sensed by a **piezoelectric crystal** (piezoelectric accelerometer - wide frequency, for vibration), **strain gauge** (piezoresistive) or **capacitive** element (MEMS accelerometer - used in air bags, phones, drones, robots).
- Natural frequency fn = (1/2 pi) sqrt(k/m); the usable range is well below fn.
- Uses: vibration monitoring, crash detection (airbag), navigation, tool

and machine condition monitoring, orientation of mobiles.

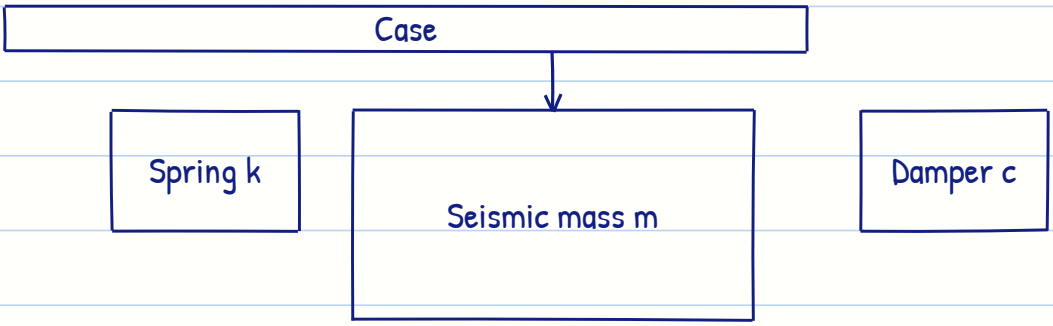


Fig: Seismic mass accelerometer

10. Level Sensors

- Measure the height of liquid or solid in a tank. **Point level** (high / low alarm) and **continuous level** sensors.
- **Float type:** a float moves with the liquid, linked to a potentiometer or a magnetic reed switch.
- **Hydrostatic pressure (differential pressure) type:** P = rho g h, so h = P / (rho g). A pressure sensor at the bottom gives the level.
- **Capacitive level sensor:** probe and tank wall form a capacitor; capacitance rises as liquid covers the probe.
- **Ultrasonic / radar:** echo time from the surface gives distance; non contact.
- **Conductive probes** (for conducting liquids), **optical** (prism) and **load cell** (weight) methods.

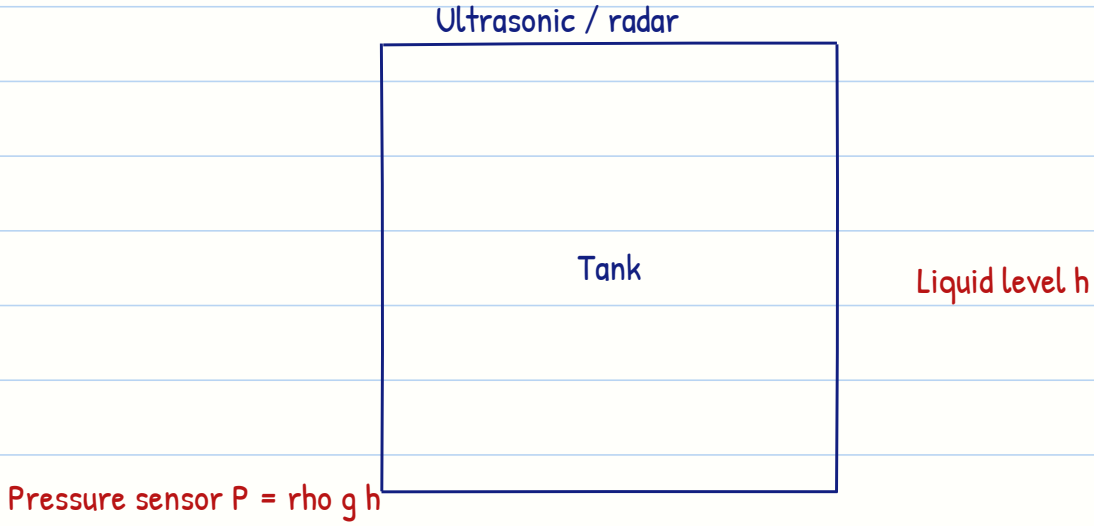


Fig: Level measurement - hydrostatic and ultrasonic methods

11. Light Sensors

- **LDR (Light Dependent Resistor):** CdS material, resistance falls when light intensity rises. Cheap; slow. Used in street lights, alarms.
- **Photodiode:** a reverse biased p-n junction; current is proportional to light. Fast, linear. **Phototransistor:** photodiode with gain.
- **Photovoltaic (solar) cell:** generates voltage from light - self generating.
- **Optical encoders, optocouplers, photo interrupters** and **CCD/CMOS image sensors** (cameras for vision systems).
- **Applications:** object counting, proximity, light curtain safety, vision inspection, bar code readers.

12. Smart Material Sensors

Smart materials change their properties with an external stimulus. They can sense and also actuate.

- **Piezoelectric materials** (quartz, PZT, PVDF): produce charge when stressed (direct effect) and strain when voltage is applied (converse effect). Sensors for force, pressure, vibration, acoustic emission.
- **Magnetostrictive materials** (Terfenol-D): change dimension in a magnetic field; used for position, torque and level sensors.
- **Shape Memory Alloys (Nitinol):** recover shape on heating; temperature sensing.
- **Fibre optic sensors:** light intensity, phase or wavelength changes with strain/temperature (Fibre Bragg Grating). Immune to EMI, good for harsh places, structural health monitoring.
- **Electro and magneto rheological fluids** , **photochromic and thermochromic** materials, **ionic polymer metal composites (IPMC)** and **conducting polymers** .
- **Hall effect and magnetoresistive** (GMR) sensors for magnetic field and speed.

13. Micro and Nano Sensors

- **Micro sensors (MEMS sensors)** are made by silicon micromachining, size 1 micron to 1 mm. Eg: MEMS accelerometer, gyroscope, pressure sensor, microphone, gas sensor, inkjet head.
- **Advantages:** small, light, low power, cheap in mass production, easy integration with electronics (smart sensor), high sensitivity.
- **Nano sensors:** using nanomaterials (**carbon nanotubes, graphene, nanowires, nanoparticles** , quantum dots) of size 1-100 nm. Large surface to volume ratio gives very high sensitivity.

- Examples: gas and chemical nanosensors, biosensors (glucose, DNA, virus), nano pressure and strain sensors, nano mechanical cantilever sensors (detects mass of single molecule).
- **Applications:** automobile (airbag, tyre pressure), medical (implants, blood pressure, glucose), consumer (mobile phone, wearables), industrial IoT, environment monitoring, defence.
- **Challenges:** packaging, noise, calibration, high cost of fabrication setup, and reliability at nano scale.

14. Selection Criteria for Sensors

- **Measurand and range:** the quantity, its minimum and maximum value.
- **Accuracy, precision, resolution** and **sensitivity** needed by the application.
- **Linearity, hysteresis, repeatability** and **drift** .
- **Dynamic response:** response time, bandwidth, natural frequency should suit the signal speed.
- **Environment:** temperature, humidity, dust, vibration, EMI, corrosive fluids, IP rating.
- **Size, weight and mounting;** contact or non-contact; **loading effect** on the measurand.
- **Output type and signal level:** voltage, current (4-20 mA), digital; compatibility with controller and signal conditioning.
- **Power supply and power consumption;** reliability and life; **cost** and availability, ease of maintenance and calibration.

Important Questions (Unit II)

- Explain static and dynamic characteristics of sensors.
- Explain the working of LVDT / encoders with neat sketches.
- Explain proximity sensors and their types.
- Explain temperature sensors - thermocouple, RTD and thermistor.
- Write short notes on force, pressure, flow, level, light and acceleration sensors.
- Write about smart material sensors, micro and nano sensors and the selection criteria for sensors.